RAJALAKSHMI ENGINEERING COLLEGE

Rakshith Y G 250701578-2025-CSE R2

Started on	Friday, 11 September 2026, 3:23 PM
State	Finished
Completed on	Friday, 11 September 2026, 3:24 PM
Time taken	29 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int arr[n];
    for (int i = 0; i < n; i++) {
        int ch;
        while ((ch = getchar()) != EOF) {
            if ((ch >= '0' && ch <= '9') || ch == '-') {
                ungetc(ch, stdin);
                break;
            }
        }
        scanf("%d", &arr[i]);
    }

    if (n == 0) {
        printf("0\n");
        return 0;
    }

    int dp[n];
    for (int i = 0; i < n; i++) {
        dp[i] = 1;
    }

    int max_len = 1;
    for (int i = 1; i < n; i++) {
        for (int j = 0; j < i; j++) {
            if (arr[i] >= arr[j]) {
                if (dp[j] + 1 > dp[i]) {
                    dp[i] = dp[j] + 1;
                }
            }
        }
        if (dp[i] > max_len) {
            max_len = dp[i];
        }
    }

    printf("%d\n", max_len);
    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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